

DATA-DRIVEN ENVIRONMENTAL ANALYSIS

RESIDENTIAL DEVELOPMENT IN PALERMO



TYPE: Environmental Analysis

YEAR: 2025

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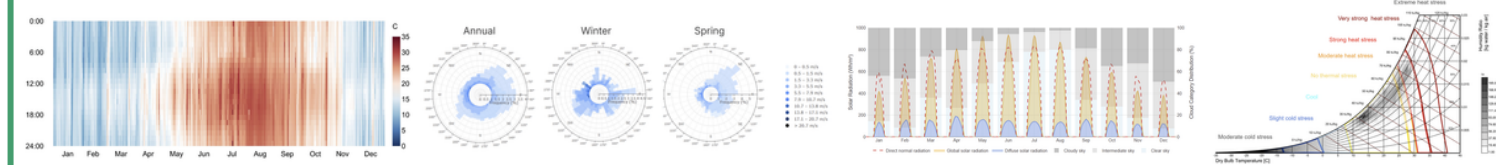
CONTEXT: IAAC Master in Advanced Computational Design (MACAD)

A performance-driven environmental analysis informs the design of a residential development and landscape in Palermo, Italy.

The study integrates climate assessment, massing placement optimization via an evolutionary algorithm based on solar radiation analysis and sea visibility, outdoor thermal comfort assessment using Honeybee, wind analysis using infrared.city, and design exploration of different voxel-based geometries.

The final proposal reduced façade solar exposure by ~30%, while enhancing daylight and sea views.

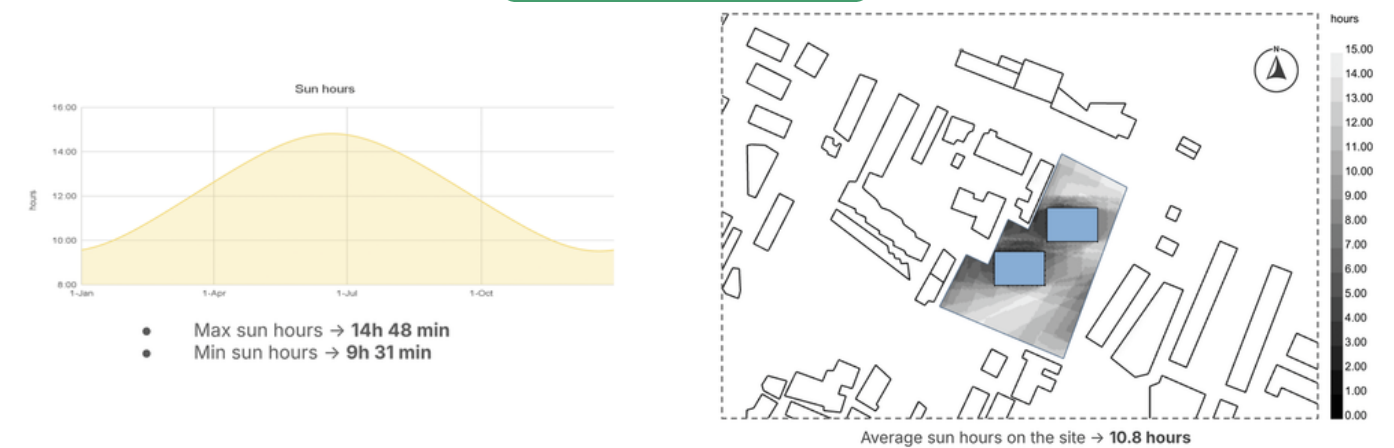
CLIMATE ANALYSIS



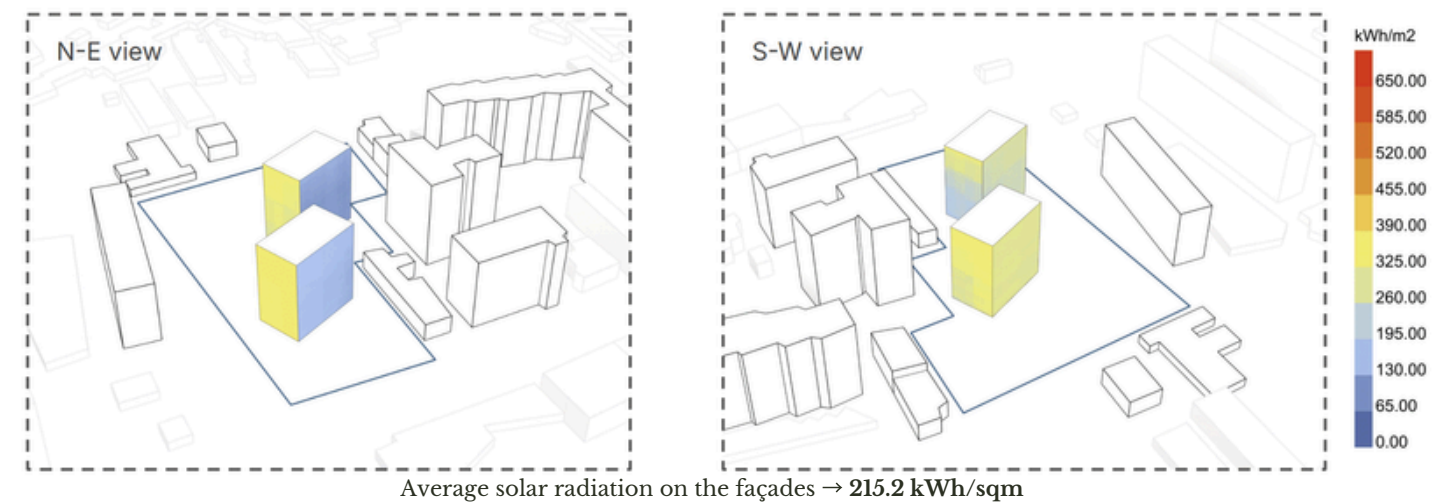
MASSING PLACEMENT OPTIMIZATION

To minimize the average solar radiation on the façades during the hottest months, the analysis was conducted with a simple box by changing the orientation, position and building dimensions. Sea visibility is increased by rotating the second building.

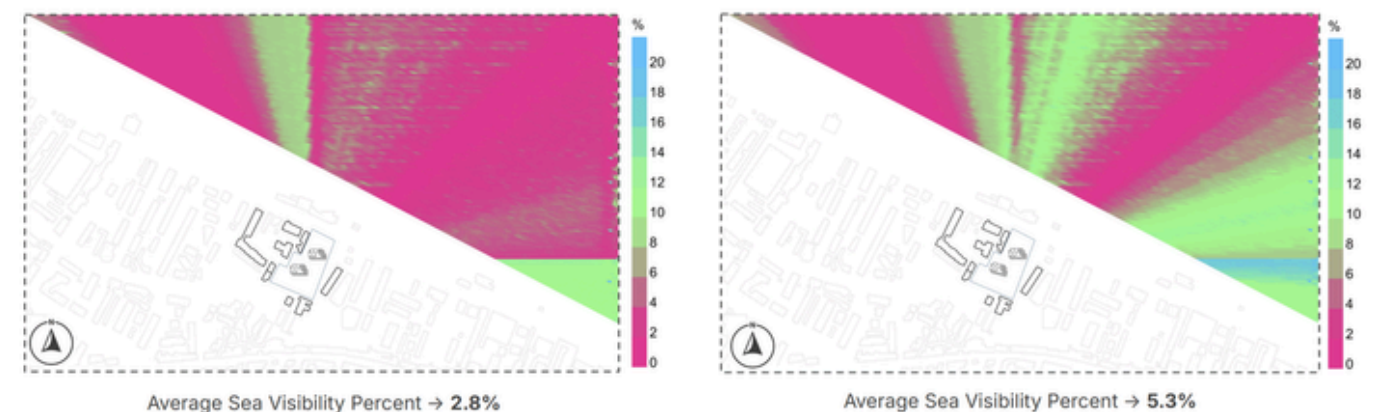
Sun Hours



Solar Radiation



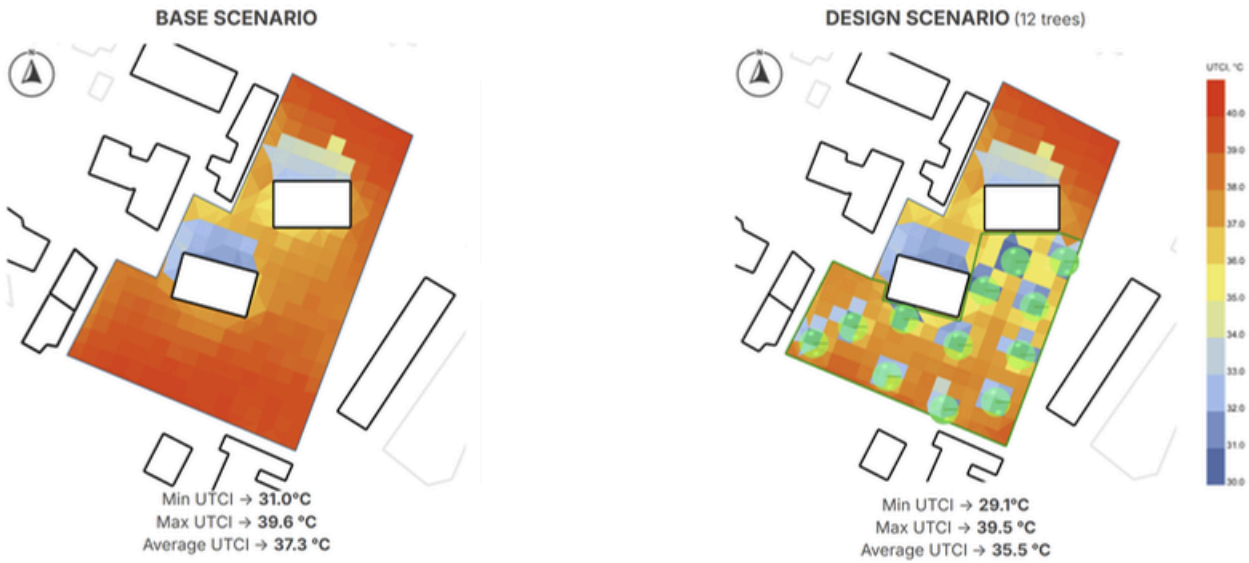
Sea Visibility



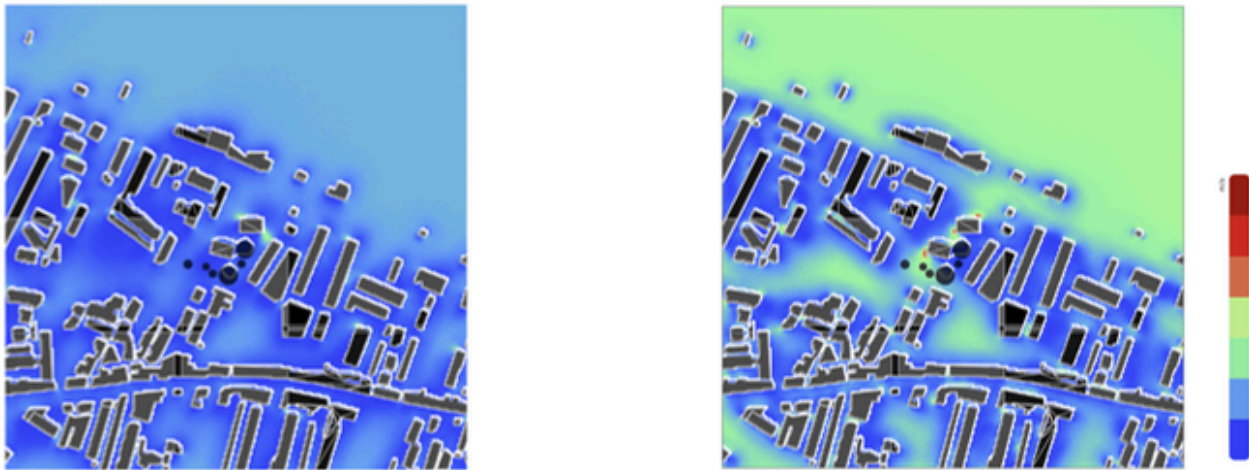
OUTDOOR COMFORT

Outdoor comfort is assessed through UTCI reduction from added trees, comparison of different ground materials and urban wind analysis.

UTCI Evaluation



Wind Analysis



Dominant Wind Speed

Max Wind Speed

DESIGN CONCEPT

SIMPLE BOX

VOXELIZATION

BALCONIES ON SOUTH FACADE



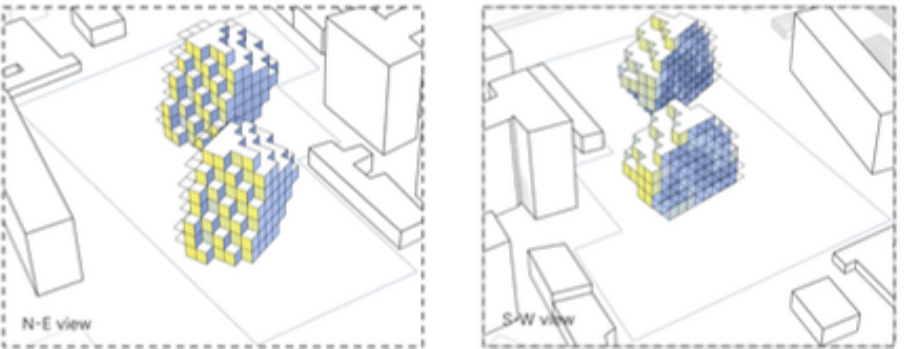
DESIGN EXPLORATION

By varying the bounding box's plan dimensions and height, different voxel-based geometries are explored and compared through a parallel coordinate chart to identify the optimal proposal.

PARAMETERS:

- Bounding box dimensions
- Volume (m³)
- Gross Floor Area (m²)
- Solar Radiation (kWh/m²)
- Daylight Factor (%)
- Visibility Percent (%)

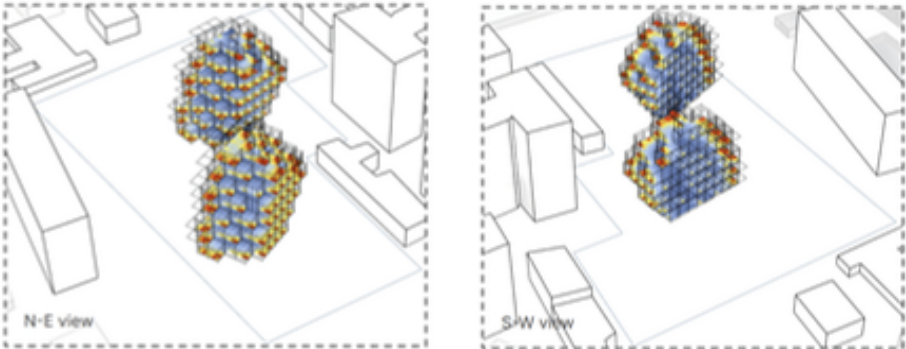
KPI:



Solar Radiation

-27.4%

compared with the simple box



Daylight Factor

+56%

compared with the 2.5% target